

1. Administrative & Study Identification

Full Study Title: An Open-label Long-term Follow-up Study to Evaluate the Effects of Sotatercept When Added to Background Pulmonary Arterial Hypertension (PAH) Therapy for the Treatment of PAH (MK-7962-038) **Short Title/Acronym:** SOTERIA / MK-7962-038 **Protocol Number:** MK-7962-038 **NCT Number:** NCT07218029 **Trial Status:** Not Yet Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Sotatercept (MK-7962) **Phase of Development:** Phase III **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of sotatercept when added to background PAH therapy in adult participants with PAH who have completed prior sotatercept studies. **Secondary Objectives:** To evaluate continued and long-term durability of efficacy in adult participants with PAH, focusing on functional and biomarker outcomes (e.g., maintaining low-risk clinical profile, Modified New York Heart Association/World Health Organization Functional Class [WHO FC], 6-minute walk distance [6MWD], and NT-proBNP levels).

2.2 Background & Rationale In Pulmonary Arterial Hypertension (PAH), the blood vessels in the lungs become thick and narrow, which makes it harder for blood to flow. This causes high pulmonary blood pressure and overworks the right heart ventricle. While standard PAH treatments alleviate symptoms, they do not halt disease progression. Sotatercept is a first-in-class biologic targeted therapy (an activin-signaling inhibitor/ActRIIA-Fc fusion protein) designed to treat PAH by working on certain proteins that play a role in causing the disease. This Long-Term Follow-Up (LTFU) study (which also formally incorporated the MK-7962-004 Amendment 07 or later / NCT04796337 study) aims to provide critical real-world data on the long-term safety, tolerability, and sustained functional improvements of sotatercept over an extended treatment period.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Long-Term Extension) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (All participants receive sotatercept)

3.2 Planned Enrollment & Duration Target \$N\$: 815 participants
Number of Sites: Global Multicenter (~255 locations) **Estimated Study Start:** 12-May-2021 **Estimated Primary Completion:** 07-Dec-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Has completed their current respective PAH sotatercept clinical study (parent study) and its requirements, and must not have discontinued early (e.g., has completed the MK-7962-004 study requirements). 3. Is willing to adhere to the study visit schedule, and understands and will comply with all protocol requirements, including providing documented informed consent. 4. Agree to use highly effective contraception methods during the study and for at least 16 weeks (112 days) after discontinuation of the study drug.

4.2 Exclusion Criteria 1. Did not participate in a sotatercept PAH parent study. 2. Missed more than the equivalent of 4 consecutive doses between the end of the parent study and the start of this study. 3. Presence of an ongoing serious adverse event (SAE) that occurred during a PAH sotatercept parent clinical study assessed to be possibly or probably related to sotatercept. 4. Is a female who is pregnant or breastfeeding. 5. Is currently enrolled in another investigational product study other than a sotatercept study (with the exception of transitioning from MK-7962-004). 6. Is or has an immediate family member who is investigational site or Sponsor staff directly involved with this study. 7. Is incapacitated.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: SC injection once every 3 weeks (Q3W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Long-term follow-up spanning up to 396 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Continued standard-of-care background PAH therapy (e.g., vasodilator combination therapy). **Prohibited:** Participation in any other trials of investigational drugs/devices for the duration of the study.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Days -14 to 0 | Informed Consent, Inclusion/Exclusion verification, Physical Exam, 12-lead ECG, WHO FC assessment, 6MWT, NT-proBNP, CPET, QoL. | | **Baseline** | Day 0 | Measurements from screening recorded as baseline values, First

Dose of Extension Phase. | | **Treatment Visits** | Every 3 Weeks | Study drug administration (SC Injection), AE/SAE Review, Concomitant Medication Review. | | **Interim Assessments** | Periodic (e.g., 24 weeks) | Rest/exercise Echocardiography, Hemoglobin/Platelets check, 6MWT, WHO FC reassessment. | | **End of Study** | Up to Week 396 | Final Safety Follow-up, Final Vital Signs, IP Accountability, Transition to commercial drug if applicable. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Number of Participants Who Experience an Adverse Event (AE):** Evaluation of long-term safety and tolerability based on any untoward medical occurrences temporally associated with the use of the study intervention. 2. **Number of Participants Who Discontinue Study Treatment Due to an AE:** Discontinuation rate resulting from adverse events. 3. **Number of Participants with Detectable Anti-Drug Antibodies (ADAs):** ADAs will be detected in serum and reported.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in 6-Minute Walk Distance (6MWD). 2. Change from baseline in N-Terminal Pro-Hormone B-type Natriuretic Peptide (NT-proBNP) levels. 3. Change from baseline in the percentage of participants who improve in Modified New York Heart Association/World Health Organization Classification of Functional Status (WHO FC). 4. Maintenance of low-risk clinical profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring temporally associated with the use of the study intervention, whether or not considered related to the study intervention, along with targeted blood samples (e.g., hemoglobin and platelets concentration) to monitor for hematological AEs. **Serious Adverse Events (SAEs):** Reported promptly per standard regulatory protocols (typically 24 hours); SAEs related to the parent study were evaluated prior to entry. **Data Monitoring Committee (DMC):** External/Internal safety monitoring committee responsible for ongoing evaluation of open-label safety signals, including ADAs and bleeding risks.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving \$1\$ dose of study treatment in the extension); Efficacy Analysis Set. **Sample Size Justification:** Target \$N=815\$ consists of a rollover population from parent studies (e.g., PULSAR, STELLAR). No formal power calculation for a primary efficacy hypothesis test is required for this open-label safety extension. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and

observed-case analysis for long-term functional shifts (Improved, No Change, Worsened WHO FC classifications).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to ensure protocol adherence, SAE reporting compliance, and proper SC administration technique. **Audit Trail:** E-CRF locking, data clarification forms, and structural data integration protocols (notably for merging data from the MK-7962-004 / NCT04796337 study records into MK-7962-038).

11. Study Identifiers & Governance

Primary ID: {{MK-7962-038}} **Registry Number:** {{NCT07218029}}

Posting ID: {{600069369413767794}} **Sponsor/Lead Organization:**

{{Merck Sharp & Dohme LLC}} **Company:** {{Merck}} **Study Title:** {{A Clinical Study of Sotatercept (MK-7962) in People With Pulmonary Arterial Hypertension (MK-7962-038)}}

Official Regulatory Title: {{An Open-label Long-term Follow-up Study to Evaluate the Effects of Sotatercept When Added to Background Pulmonary Arterial Hypertension (PAH) Therapy for the Treatment of PAH (MK-7962-038)}}

12. Clinical Framework (Adapted for PAH)

Primary Condition: {{Pulmonary Arterial Hypertension (PAH)}}

Disease Area: {{Pulmonary Arterial Hypertension}} **Diagnosis**

Threshold: {{Confirmed WHO Group 1 PAH & completion of a parent sotatercept trial}}

Medical Methodology: {{Biologic targeted therapy (Sotatercept) added to standard background PAH vasodilator therapy}}

Optimization Target: {{Sustained restoration of activin-BMP signaling balance / Maintenance of low-risk WHO FC profile}}

13. Study Procedures & Methodology

Management Pathway: {{Long-Term Follow-Up (LTFU) extension clinical pathway for PAH}}

Education Protocol (HCP): {{Site training on long-term biologic SC administration, hematologic monitoring, and ADA tracking}}

Education Protocol (Patient): {{Education on SC injection site care, signs of PAH progression, and prompt AE/bleeding reporting}}

Quality Audit Mechanism: {{Routine safety data review, tracking of WHO FC evaluation consistency, and parent-to-extension rollover reconciliation}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 396 weeks (LTFU)}} **Onsite Visit Cadence:** {{Every 3 weeks for treatment administration and routine vitals}} **Remote Monitoring Frequency:** {{Standard interval monitoring per Sponsor Data Management Plan}} **Checklist Review Interval:** {{Periodic safety and functional re-evaluations (e.g., 6MWD, ECHO, NT-proBNP) at predefined milestones}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					